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Miller, II

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(54) **SYSTEM AND METHOD OF OPTICAL**
ANTENNA TUNING

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- (71) Applicant: **AT&T Intellectual Property I, L.P.**,
Atlanta, GA (US)
- (72) Inventor: **Robert R. Miller, II**, Convent Stn., NJ
(US)
- (73) Assignee: **AT&T Intellectual Property I, L.P.**,
Atlanta, GA (US)
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Primary Examiner — Awat M Salih

(74) *Attorney, Agent, or Firm* — Guntin & Gust, PLC;
Dana B. LeMoine

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H04B 10/80 (2013.01)

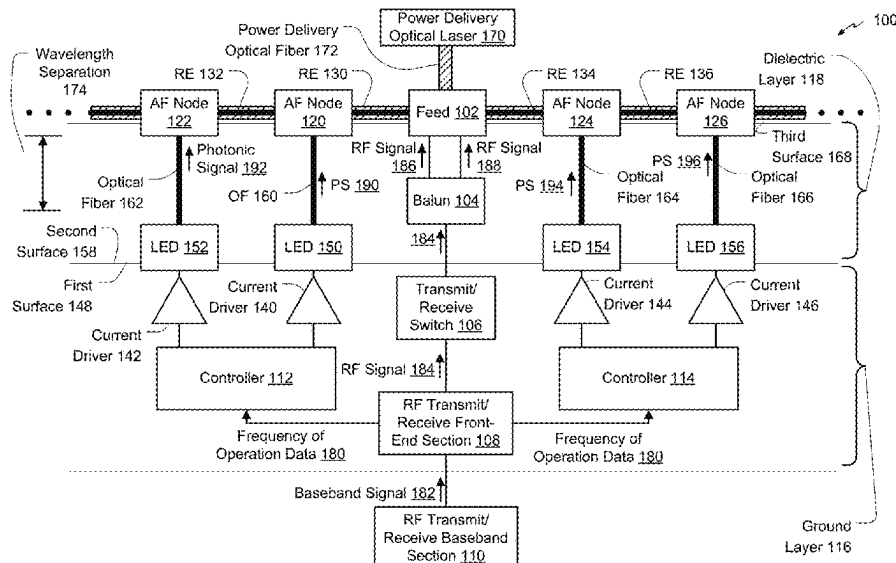
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(57) **ABSTRACT**

A method includes determining an operating frequency of an antenna at a component of the antenna. The method includes determining, at the component of the antenna, a filter node of a plurality of filter nodes to activate to tune the antenna to the operating frequency. The method includes transmitting power to the filter node, wherein the power is transmitted via a first optical fiber. The method also includes sending a signal from the component to a light source. The activation of the light source sends an optical signal to the filter node. The filter node adjusts a characteristic of a radiating element coupled to the filter node using the power responsive to the optical signal. Adjustment of the characteristic facilitates tuning the antenna to the operating frequency.

20 Claims, 4 Drawing Sheets



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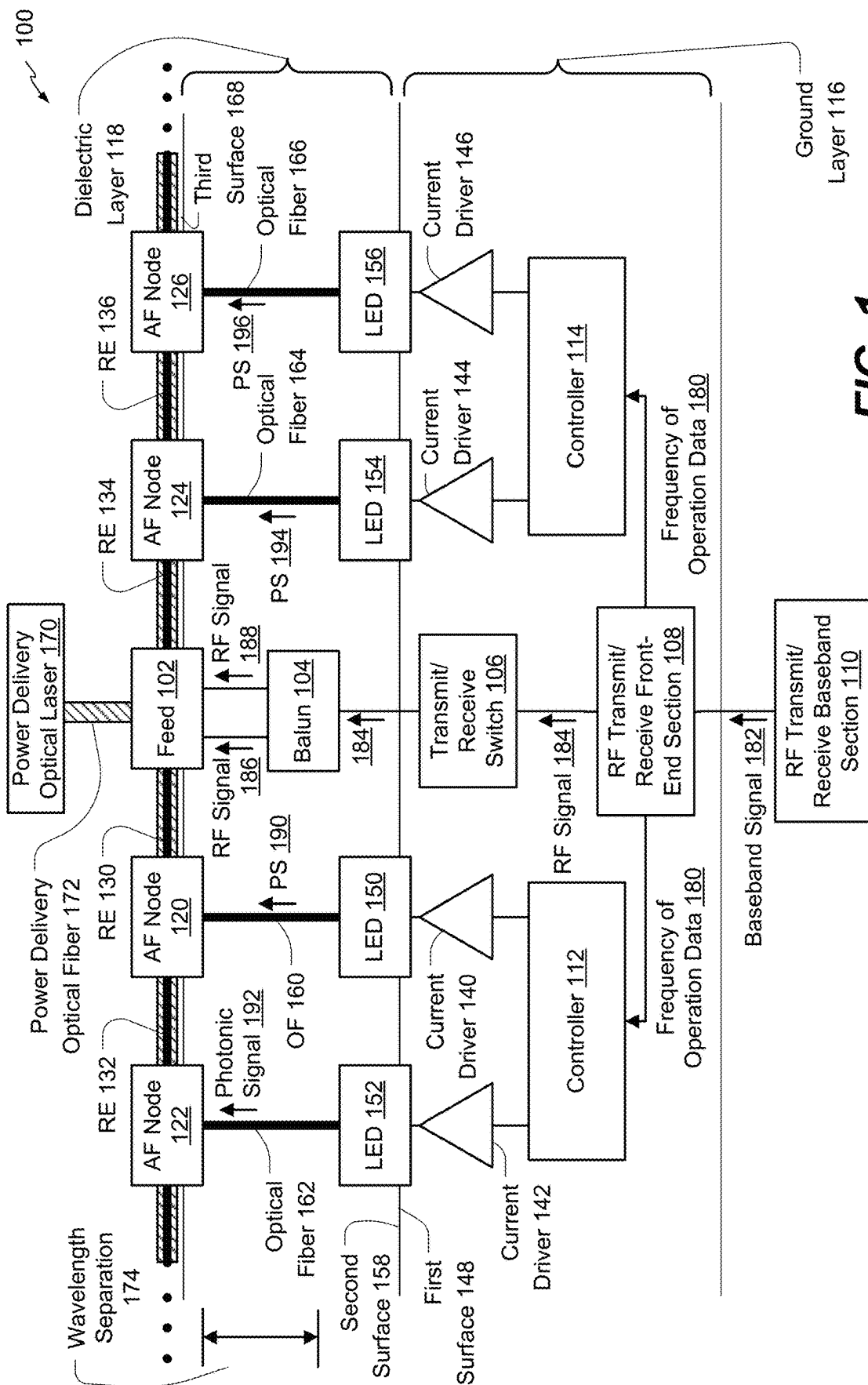


FIG. 1

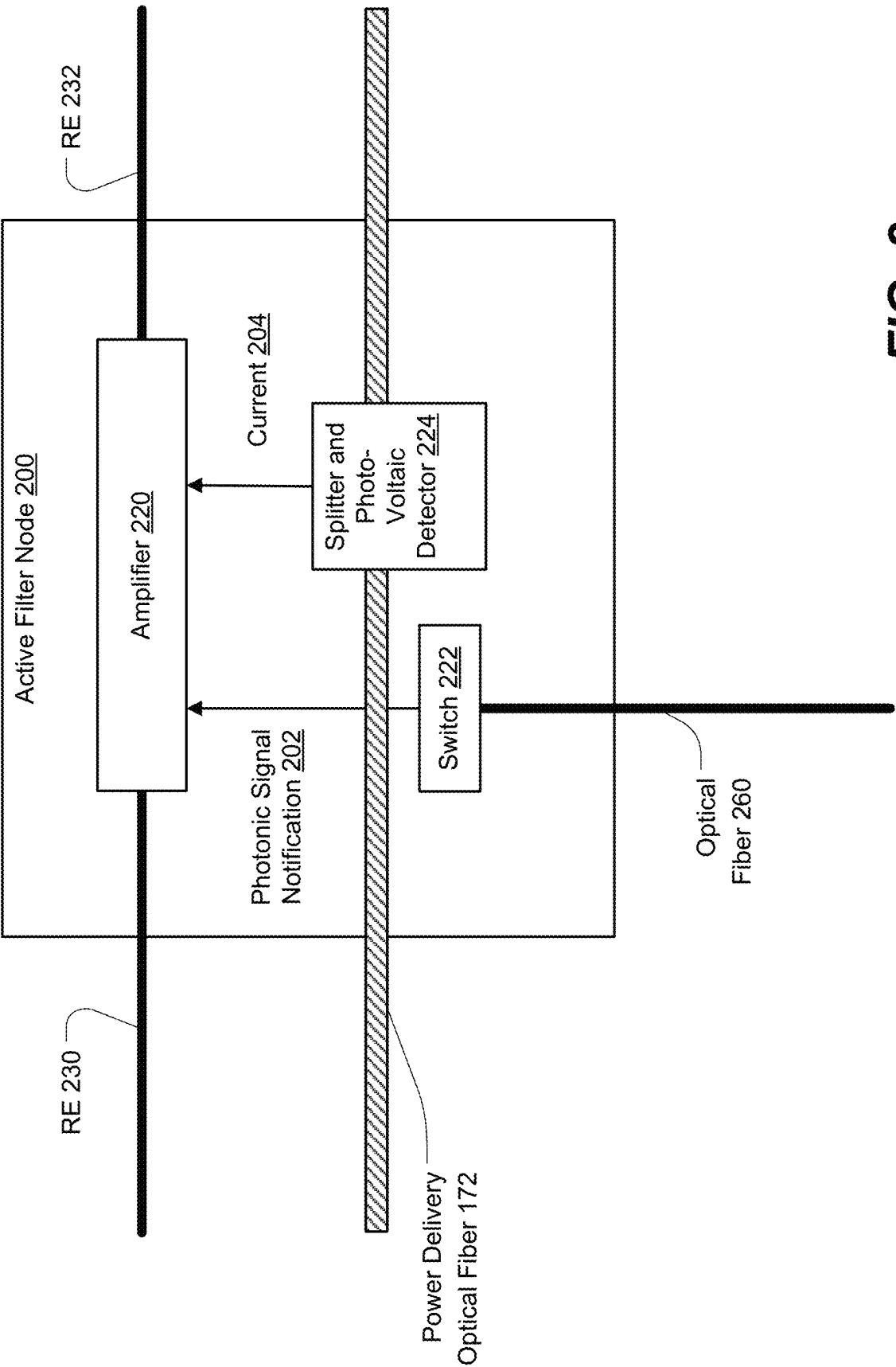


FIG. 2

↖ 300

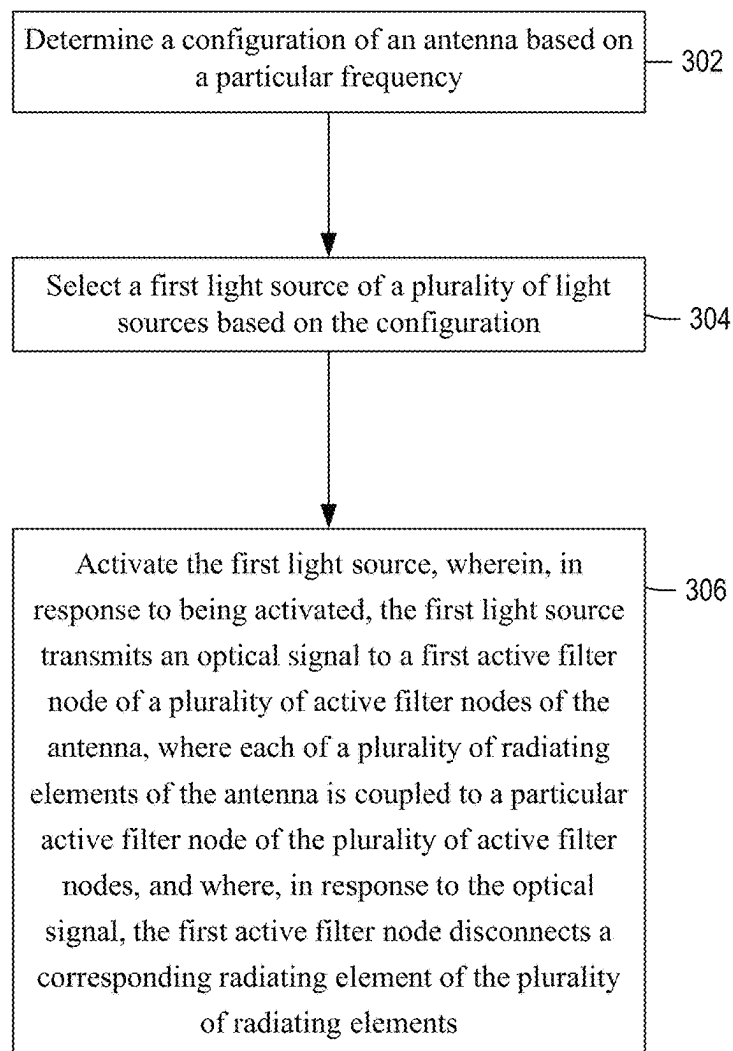
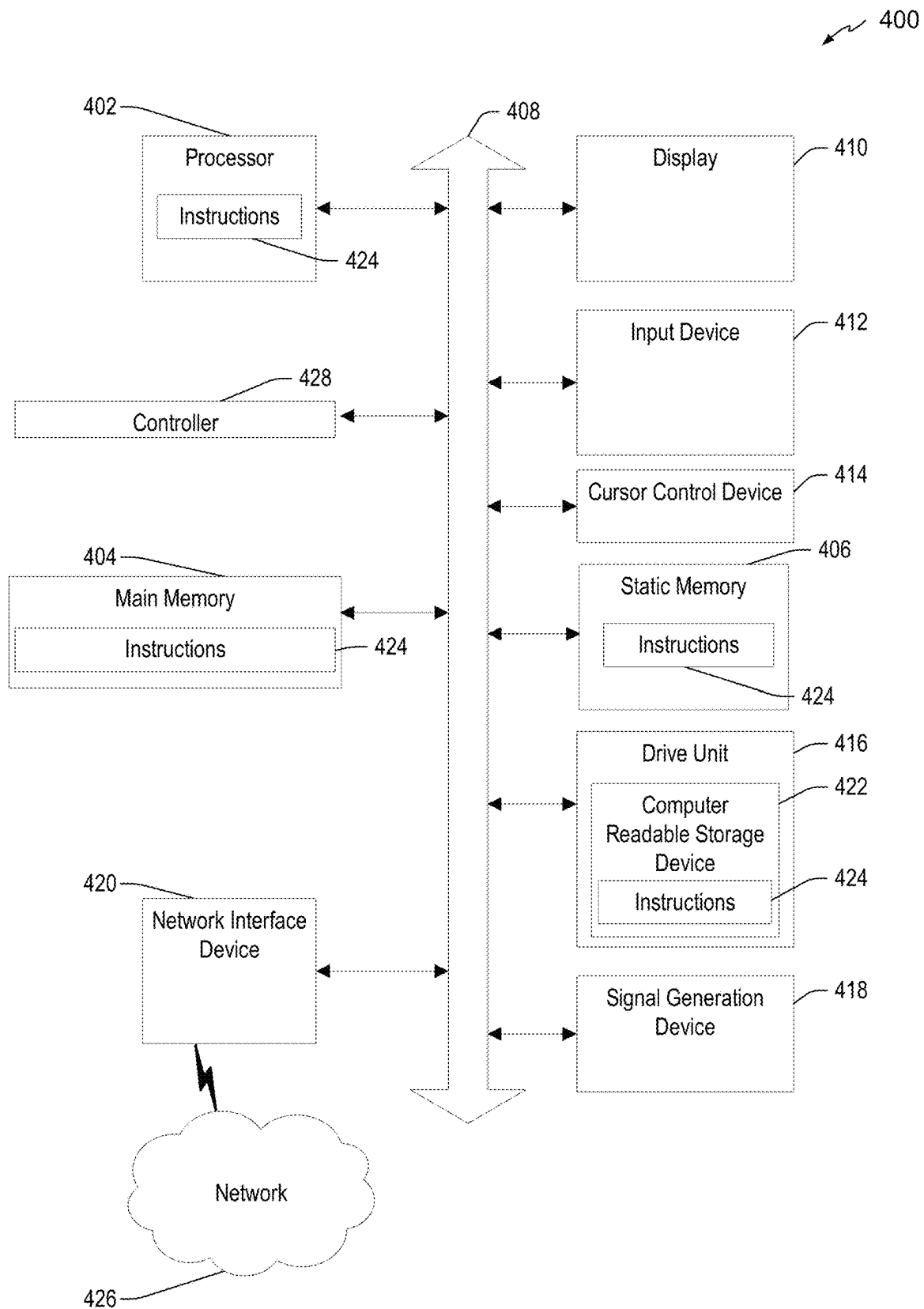


FIG. 3



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SYSTEM AND METHOD OF OPTICAL ANTENNA TUNING

CROSS REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 17/499,442, filed on Oct. 12, 2021, which is a divisional of U.S. patent application Ser. No. 15/977,169, filed on May 11, 2018 (now U.S. Pat. No. 11,177,574), which is a continuation of U.S. patent application Ser. No. 14/084,200, filed Nov. 19, 2013 (now U.S. Pat. No. 10,003,131), which are incorporated herein by reference in their entirety.

FIELD OF THE DISCLOSURE

The present disclosure is generally related to optical antenna tuning.

BACKGROUND

Antennas are used as coupling elements between circuits (e.g., radio frequency (RF) circuits) and a wireless medium. Antenna design usually uses metallic conductors to transfer energy between electromagnetic waves and electrical currents. An antenna's size is usually related to a target wavelength. For example, an antenna may have a characteristic dimension (e.g., length) that is $\frac{1}{2}\lambda$ or $\frac{1}{4}\lambda$ of a target wavelength.

Some antennas may be used to operate over a wide operating frequency band. Fixed tuning mechanisms, such as frequency-selective lumped filters (called "traps"), may be used to create multi-band antennas that selectively add reactance to portions of a longer antenna to allow the antenna to operate in frequency bands with shorter wavelengths. Reactance may be added to the portions of the antenna by joining segments of a metallic radiating structure of the antenna using traps or filters. Multi-band antennas may also be created using distributed resonant (LC) element structures.

When a spectral coverage of an antenna is very large, it may be useful to have an antenna that is "active" in the sense that metallic element segments of the antenna may be adjusted remotely. One approach has been to include an extendable telescoping mast in the antenna. A length of the telescoping mast may be varied by using a motor drive system. Using an extendable telescoping mast for an antenna in the microwave region may be impractical, as the antenna may be small and the adjustments may distort the behavior of the antenna.

Another approach to adjust an antenna such that a radiating length is matched to a frequency of interest is to use electrical switches or relays. Control and power circuits used to provide adjustment may disrupt the radiative behavior of the antenna due to parasitic conductors that behave as unwanted radiating elements.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a block diagram of a particular illustrative embodiment of an antenna that may be optically tuned;

FIG. 2 is a diagram of an illustrative embodiment of an active filter node that may be included in the antenna of FIG. 1;

FIG. 3 is a flowchart to illustrate a particular embodiment of a method of optically tuning the antenna of FIG. 1; and

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FIG. 4 is a block diagram of an illustrative embodiment of a general computer system operable to support embodiments of computer-implemented methods, computer program products, and system components as illustrated in FIGS. 1-3.

DETAILED DESCRIPTION

Systems and methods of optically tuning an antenna are disclosed. Optical signals (e.g., photonic signals) may be used to actively tune the antenna using an electronic switching device (such as a photosensitive field effect transistor). Metallic radiating segments that form the antenna may be selectively activated or deactivated to tune the antenna to a target frequency of operation. The optical signals may be supplied by a light emitting diode (LED), a laser, or another source. The optical signals may be conveyed to an electronic switching device via a fiber optic cable, a light-guide, or via free-space. The optical signal source may be separated from the antenna by a sufficiently long distance to prevent a field of the antenna from being affected by the optical signal source.

In a particular embodiment, distributed amplifiers may operate as adjustable filters. The distributed amplifiers may adjust parameters (e.g., inductance, capacitance, and/or gain) of the antenna segments. Power-over-fiber (POF) techniques may be used to power the amplifiers. Additionally, optical fibers may be used to provide control signals to the amplifiers. Moreover, the disclosed techniques may be extensible to mass/miniaturized manufacturing techniques.

In a particular embodiment, a method includes determining a configuration of an antenna. The method also includes selecting a first light source of a plurality of light sources based on the configuration. The method further includes activating the first light source and transmitting an optical signal to a first active filter node of a plurality of active filter nodes of the antenna. In response to receiving the optical signal, the first active filter node deactivates a corresponding radiating element of a plurality of radiating elements.

In another particular embodiment, an antenna includes a plurality of active filter nodes and a plurality of radiating elements. Each of the plurality of radiating elements is coupled to a particular active filter node of the plurality of active filter nodes. The antenna also includes a plurality of light sources and a controller. The controller activates a first light source of the plurality of light sources. The first light source, in response to being activated, transmits an optical signal to a first active filter node of the plurality of active filter nodes. The first active filter node deactivates a corresponding radiating element of the plurality of radiating elements in response to receiving the optical signal.

In another particular embodiment, a computer-readable storage device stores instructions that, when executed by a processor, cause the processor to perform operations including determining a configuration of an antenna based on a particular frequency, selecting a first light source of a plurality of light sources based on the configuration, and activating the first light source. The first light source, in response to being activated, transmits an optical signal to a first active filter node of a plurality of active filter nodes of the antenna. The first active filter node deactivates (or adjusts) a corresponding radiating element of a plurality of radiating elements in response to receiving the optical signal.

Referring to FIG. 1, a particular illustrative embodiment of an antenna is disclosed and generally designated **100**. The antenna **100** may include a plurality of radiating elements

(e.g., radiating elements 130, 132, 134, and 136). One or more of the plurality of radiating elements 130, 132, 134, and 136 may be metallic. The antenna 100 may also include a plurality of active filter nodes (e.g., active filter nodes 120, 122, 124, and 126). A particular illustrative embodiment of an active filter node is described with reference to FIG. 2.

Each of the radiating elements may be coupled to a particular active filter node. For example, the radiating element 130 may be coupled to the active filter node 120, the radiating element 132 may be coupled to the active filter node 122, the radiating element 134 may be coupled to the active filter node 124, and the radiating element 136 may be coupled to the active filter node 126.

The antenna 100 may include a radio frequency (RF) transmit/receive baseband section 110 coupled to an RF transmit/receive front-end section 108. The RF transmit/receive front-end section 108 may be coupled to a feed 102 via a transmit/receive switch (or a duplex filter) 106 and via a balun 104. The RF transmit/receive front-end section 108 may be coupled to one or more controllers (e.g., controllers 112 and 114). In a particular embodiment, the antenna 100 may correspond to a monopole antenna.

Each of the controllers 112 and 114 may be coupled to one or more light sources, such as light emitting diodes (LEDs) (e.g., LEDs 150, 152, 154, and 156). For example, the controller 112 may be coupled to the LED 150 via a current driver 140 and may be coupled to the LED 152 via a current driver 142. As another example, the controller 114 may be coupled to the LED 154 via a current driver 144 and may be coupled to the LED 156 via a current driver 146. In a particular embodiment, one or more of the LEDs (e.g., the LEDs 150, 152, 154, and 156) may be replaced by a laser, or another light source.

In a particular embodiment, each of the light sources may be coupled, via an optical fiber, to a corresponding active filter node. For example, the LED 150 may be coupled, via an optical fiber 160, to the active filter node 120. The LED 152 may be coupled, via an optical fiber 162, to the active filter node 122. The LED 154 may be coupled, via an optical fiber 164, to the active filter node 124. The LED 156 may be coupled, via an optical fiber 166, to the active filter node 126. An optical fiber may also be referred to as a fiber optic cable. In another embodiment, the LEDs 150, 152, 154, 156 may be separated from the active filter nodes 120, 122, 124, 126 by free-space.

In a particular embodiment, the feed 102 may be coupled to a power delivery optical laser 170 via a power delivery optical fiber 172. In this embodiment, each of the active filter nodes 120, 122, 124, and 126 may be coupled to the power delivery optical fiber 172. The feed 102 may be powered by the power delivery optical laser 170 via the power delivery optical fiber 172. In a particular embodiment, one or more of the active filter nodes 120, 122, 124, 126 may be powered by the power delivery optical laser 170 via the power delivery optical fiber 172, as further described with reference to FIG. 2.

A distance (e.g., a wavelength separation 174) between each of the active filter nodes 120, 122, 124, or 126 and a corresponding LED 150, 152, 154, or 156 may be greater than a threshold distance. The wavelength separation 174 may be sufficient such that a field of the antenna 100 is unaffected (or substantially unaffected) by the LEDs 150, 152, 154, and 156 and associated drive circuits. In a particular embodiment, a frequency of operation of the antenna 100 may be based on a number of radiating elements (e.g., radiating elements 130, 132, 134, and 136) that are activated (or tuned).

During operation, the controllers 112, 114 may deactivate one or more of the radiating elements 130, 132, 134, and 136 based on a particular frequency (e.g., megahertz (MHz) or gigahertz (GHz)) of operation. For example, the controller 112 may deactivate one of the radiating elements 130 and 132 based on a first frequency of operation, may deactivate both of the radiating elements 130 and 132 based on a second frequency of operation, and may deactivate neither of the radiating elements 130 and 132 based on a third frequency of operation. In a particular embodiment, the controllers 112, 114 may select radiating elements that are farthest to the feed 102 to achieve the particular frequency of operation. The controllers 112, 114 may receive frequency of operation data 180 from the RF transmit/receive front-end section 108. The frequency of operation data 180 may indicate the particular frequency of operation to which the antenna 100 is to be tuned.

The controllers 112, 114 may deactivate one or more of the radiating elements 130, 132, 134, and 136 using one or more of the LEDs 150, 152, 154, and 156. For example, the controllers 112, 114 may activate a single light source or multiple light sources of the one or more light sources (e.g., the LEDs 150, 152, 154, or 156). To illustrate, the controller 112 may activate the LED 150 using the current driver 140, may activate the LED 152 using the current driver 142, or both. When activated, each of the LEDs 150, 152, 154, and 156 may transmit an optical signal (e.g., a photonic signal 190, 192, 194, or 196) to one or more corresponding active filter nodes 120, 122, 124, and 126. In response to receiving the optical signal (e.g., the photonic signal 190, 192, 194, or 196), each of the corresponding active filter nodes 120, 122, 124, and 126 may deactivate a corresponding radiating element 130, 132, 134, or 136.

In a particular embodiment, each of the one or more corresponding active filter nodes 120, 122, 124, and 126 may activate a corresponding radiating element 130, 132, 134, or 136 in response to the optical signal (e.g., the photonic signal 190, 192, 194, or 196). In this embodiment, each of the corresponding one or more active filter nodes 120, 122, 124, and 126 may deactivate the corresponding radiating element 130, 132, 134, or 136 based on not receiving the optical signal (e.g., the photonic signal 190, 192, 194, or 196) for a threshold duration. For example, the active filter node 120 may receive an optical signal (e.g., the photonic signal 190) at a first time. The active filter node 120 may activate the radiating element 130 in response to receiving the optical signal (e.g., the photonic signal 190). Subsequently, the active filter node 120 may deactivate the radiating element 130 in response to not receiving another optical signal (e.g., the photonic signal 190) within a threshold duration of the first time.

In a particular embodiment, a particular active filter node may operate on (e.g., deactivate, activate, disconnect, connect, partially disconnect, partially connect, or change a reactance of) a single radiating element. For example, the particular active filter node (e.g., the active filter node 120) may be coupled to two radiating elements (e.g., the radiating elements 130 and 132). In response to receiving the photonic signal 190, the active filter node 120 may deactivate only the radiating element 132. In a particular embodiment, in response to receiving the photonic signal 190 the active filter node 120 may deactivate only the radiating element 130.

The optical signal (e.g., the photonic signal 190, 192, 194, or 196) may be transmitted from an LED 150, 152, 154, 156 to a corresponding active filter node 120, 122, 124, 126 via the corresponding optical fiber 160, 162, 164, 166. In a particular embodiment, the optical signal (e.g., the photonic

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signal 190, 192, 194, or 196) may be transmitted from the corresponding LED 150, 152, 154, 156 to the corresponding active filter node 120, 122, 124, 126 via free-space.

While operating in a transmit mode of operation, the RF transmit/receive baseband section 110 may transmit a baseband signal 182 to the RF transmit/receive front-end section 108. The RF transmit/receive front-end section 108 may generate a radio frequency (RF) signal 184 based on the baseband signal 182. The transmit/receive switch 106 may be set to transmit in the transmit mode of operation. The RF transmit/receive front-end section 108 may transmit the RF signal 184 to the balun 104 via the transmit/receive switch 106. The balun 104 may be used to match the antenna 100 or the one or more active filter nodes 120, 122, 124, and 126. In a particular embodiment, the antenna 100 may correspond to a dipole antenna. In a particular embodiment, the one or more active filter nodes 120, 122, 124, and 126 may include a single-ended switch, a duplexer filter, or both. The balun 104 may generate a plurality of signals (e.g., RF signals 186 and 188) based on the RF signal 184. The balun 104 may transmit the RF signals 186 and 188 to the feed 102.

The feed 102 may transmit the RF signals 186 and 188 to one or more of the radiating elements 130, 132, 134, and 136. For example, the feed 102 may transmit the RF signal 186 to the radiating elements 130 and 132 on one side of the feed 102 and may transmit the RF signal 188 to the radiating elements 134 and 136 on the other side of the feed 102. The activated radiating element(s) of the radiating elements 130 and 132 may transmit the RF signal 186 over a wireless medium. For example, the activated radiating element(s) of the radiating elements 130 and 132 may radiate energy from the RF signal 186 as electromagnetic waves (radio waves) over the wireless medium. Similarly, the activated radiating element(s) of the radiating elements 134 and 136 on the other side of the feed 102 may transmit the RF signal 188 over the wireless medium. The particular frequency of operation to which the antenna 100 is tuned may be based on how many of the radiating elements 130, 132, 134, and 136 are activated. For example, a higher number (e.g., 4) of activated radiating element(s) of the radiating elements 130, 132, 134, and 136 may correspond to a longer antenna and a lower (e.g., 0.5 GHz) frequency of operation. As another example, a lower number (e.g., 2) of activated radiating element(s) of the radiating elements 130, 132, 134, and 136 may correspond to a shorter antenna and a higher (e.g., 1 GHz) frequency of operation. The RF signals 186 and 188 may be transmitted by the activated radiating element(s) of the radiating elements 130, 132, 134, and 136 at the particular frequency of operation to which the antenna 100 is tuned.

In a particular embodiment, the antenna 100 may include a ground layer (e.g., a ground layer 116) aligned with a dielectric layer (e.g., a dielectric layer 118). For example, the dielectric layer 118 may be between the ground layer 116 and the radiating elements 130, 132, 134, and 136. The optical signal (e.g., the photonic signals 190, 192, 194, or 196) may be transmitted from the one or more light sources (e.g., the LEDs 150, 152, 154, or 156) through the dielectric layer 118 to corresponding active filter nodes (e.g., the active filter nodes 120, 122, 124, or 126). In a particular embodiment, each of the one or more light sources (e.g., the LEDs 150, 152, 154, or 156) may be located on (or embedded in) a surface (e.g., a first surface 148) of the ground layer 116. The first surface 148 may be adjacent to a surface (e.g., a second surface 158) of the dielectric layer 118. Each of the plurality of active filter nodes (e.g., the active filter nodes 120, 122, 124, or 126) may be located on another surface

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(e.g., a third surface 168) of the dielectric layer 118. The second surface 158 and the third surface 168 may be parallel to each other and may be on opposite sides of the dielectric layer 118. To illustrate, the photonic signal 190 may be transmitted from the LED 150 through the dielectric layer 118 to the active filter node 120. Mounting the plurality of active filter nodes (e.g., the active filter nodes 120, 122, 124, or 126) on the dielectric layer 118 may reduce the threshold distance of the wavelength separation 174, decreasing the overall size of the antenna 100.

In a particular embodiment, the optical signal (e.g., the photonic signals 190, 192, 194, or 196) may be transmitted from the one or more light sources (e.g., the LEDs 150, 152, 154, or 156) through at least a portion of the ground layer 116 prior to being transmitted through the dielectric layer 118 to the corresponding active filter nodes (e.g., the active filter nodes 120, 122, 124, or 126). For example, the photonic signal 190 may be transmitted from the LED 150 through at least a portion of the ground layer 116 prior to being transmitted through the dielectric layer 118 to the active filter node 120. The one or more light sources (e.g., the LEDs 150, 152, 154, or 156) may be located within the ground layer 116, on (or embedded in) a surface of the ground layer 116, outside the ground layer 116, or a combination thereof.

In a particular embodiment, the optical signal (e.g., the photonic signals 190, 192, 194, or 196) is transmitted via one or more light guides. For example, the photonic signal 190 may be transmitted from the LED 150 through the dielectric layer 118 via a light guide to the active filter node 120. As another example, the photonic signal 190 may travel from the LED 150 through at least a portion of the ground layer 116 via a light guide prior to being transmitted through the dielectric layer 118 to the active filter node 120.

In a particular embodiment, the optical signal (e.g., the photonic signals 190, 192, 194, or 196) travels in a substantially straight path from the one or more light sources (e.g., the LEDs 150, 152, 154, or 156) to the corresponding active filter nodes (e.g., the active filter nodes 120, 122, 124, or 126). In an alternative embodiment, the optical signal (e.g., the photonic signals 190, 192, 194, or 196) travels in a curved (or angular) path from the one or more light sources (e.g., the LEDs 150, 152, 154, or 156) to the corresponding active filter nodes (e.g., the active filter nodes 120, 122, 124, or 126). For example, the photonic signal 190 may travel via a curved light guide through at least a portion of the ground layer 116, the dielectric layer 118, or both.

In a particular embodiment, the ground layer 116 includes at least a portion of one or more components of the antenna 100. For example, the ground layer 116 may include at least a portion of one or more of the current drivers 140, 142, 144, 146, the controllers 112, 114, the transmit/receive switch 106, the RF transmit/receive front-end section 108, the transmit/receive baseband section 110, the balun 104, the one or more light sources (e.g., the LEDs 150, 152, 154, or 156), the feed 102, the plurality of active filter nodes (e.g., the active filter nodes 120, 122, 124, or 126), or the power delivery optical laser 170.

In a particular embodiment, the dielectric layer 118 may include at least a portion of one or more components of the antenna 100. For example, the dielectric layer 118 may include at least a portion of one or more of the current drivers 140, 142, 144, 146, the controllers 112, 114, the transmit/receive switch 106, the RF transmit/receive front-end section 108, the transmit/receive baseband section 110, the balun 104, the one or more light sources (e.g., the LEDs 150, 152, 154, or 156), the feed 102, the plurality of active

filter nodes (e.g., the active filter nodes **120**, **122**, **124**, or **126**), or the power delivery optical laser **170**.

The controllers **112**, **114** may deactivate one or more of the radiating elements **130**, **132**, **134**, **136** using optical signals based on a particular frequency of operation of the antenna **100**. Having the dielectric layer **118** between the one or more light sources (e.g., the LEDs **150**, **152**, **154**, or **156**) and the corresponding active filter nodes (e.g., the active filter nodes **120**, **122**, **124**, or **126**) may reduce the overall size of the antenna **100**, making the antenna **100** suitable for miniaturized manufacturing techniques.

Referring to FIG. 2, an active filter node is disclosed and generally designated **200**. In a particular embodiment, the active filter node **200** corresponds to one of the active filter nodes **120**, **122**, **124**, or **126** of FIG. 1. The active filter node **200** may include a capacitively-coupled radio frequency (RF) bi-directional active filter node. The active filter node **200** includes an amplifier **220** coupled to a radiating element **230** on one side of the amplifier **220** and to a radiating element **232** on another side of the amplifier **220**. In a particular embodiment, each of the radiating elements **230** and **232** corresponds to one of the radiating elements **130**, **132**, **134**, and **136** of FIG. 1. In a particular embodiment, the amplifier **220** may include a capacitively-coupled amplifier, a bi-directional amplifier, a radio frequency amplifier, or any combination thereof. The active filter node **200** may, for example, include a Bessel filter transfer function. The active filter node **200** may implement a bi-quadratic transfer function which may synthesize other filters. The active filter node **200** may include a splitter, a photo-voltaic detector, a rectifying photo-voltaic detector, or a combination thereof (e.g., a splitter and photo-voltaic detector **224**). The active filter node **200** may be coupled to the power delivery optical fiber **172** via the splitter and photo-voltaic detector **224**. In a particular embodiment, the splitter and photo-voltaic detector **224** may include a splitter and a rectifying photo-voltaic detector. The active filter node **200** may include a switch **222**. In a particular embodiment, the switch **222** may include a bi-polar transistor, a field effect transistor, an analog photo-responsive switch, an analog photo-responsive resistor, or a combination thereof. The switch **222** may be coupled to an optical fiber **260**. In a particular embodiment, the optical fiber **260** corresponds to one of the optical fibers **160**, **162**, **164**, or **166** of FIG. 1.

During operation, the amplifier **220** may receive a photonic signal notification **202** from the switch **222**. For example, the switch **222** may transmit the photonic signal notification **202** to the amplifier **220** in response to receiving a photonic signal (e.g., the photonic signal **190**, **192**, **194**, or **196** of FIG. 1). In a particular embodiment, the active filter node **200** corresponds to the active filter node **120** of FIG. 1. In this embodiment, the switch **222** transmits the photonic signal notification **202** to the amplifier **220** in response to receiving the photonic signal **190** of FIG. 1. The amplifier **220** deactivates one or more of the radiating element **230**, **232** in response to receiving the photonic signal notification **202**. In another particular embodiment, the amplifier **220** may activate one or more of the radiating element **230**, **232**, in response to receiving the photonic signal notification **202**.

In a particular embodiment, the amplifier **220** is powered by the splitter and photo-voltaic detector **224**. For example, the splitter and photo-voltaic detector **224** may generate a current **204** based on detecting transmission of light through the power delivery optical fiber **172** and may transmit the current **204** to the amplifier **220**.

Thus, the antenna **100** may include the active filter node **200** with the amplifier **220**. Use of the active filter node **200**

may replace use of trap in the antenna **100**. The active filter node **200** may enable the antenna **100** to operate at a higher frequency bandwidth than the trap.

FIG. 3 is a flowchart to illustrate a particular embodiment of a method **300** of optically tuning an antenna. In an illustrative embodiment, the method **300** may be performed to tune the antenna **100** of FIG. 1.

The method **300** includes determining a configuration of an antenna based on a particular frequency, at **302**. For example, in FIG. 1, the controllers **112**, **114** may determine a configuration of the antenna **100** based on a particular frequency using frequency and configuration correlation data. To illustrate, the frequency and configuration correlation data may map specific frequencies (e.g., 1 GHz) to a particular number (e.g., 2) of deactivated (or activated) radiating elements of the plurality of radiating elements (e.g., the radiating elements **130**, **132**, **134**, and **136**. Each of the radiating elements **130**, **132**, **134**, and **136** may have an equivalent effect on the frequency of operation of the antenna **100**.

For example, a frequency range (e.g., spanning 500 MHz to 2 GHz) may correspond to a particular dipole overall antenna length (e.g., approximately 12 inches). Each of the radiating elements **130**, **132**, **134**, and **136** may correspond to a particular antenna length portion (e.g., 3 inches) when activated. A number (e.g., $12/3=4$) of activated radiating elements corresponding to a specific frequency (e.g., 500 MHz) may be based on the particular overall antenna length (e.g., approximately 12 inches) and the particular antenna length portion (e.g., 3 inches). The number (e.g., $4-4=0$) of deactivated radiating elements corresponding to the specific frequency (e.g., 500 MHz) may be based on the number (e.g., 4) of activated radiating elements and a total number (e.g., 4) of radiating elements of the antenna **100**.

In a particular embodiment, any of the radiating elements **130**, **132**, **134**, and **136** adding up to the particular number are selectable by the controllers **112**, **114** to achieve the particular frequency. In an alternative embodiment, the particular number of radiating elements that are farthest (or nearest) to the feed **102** are selectable by the controllers **112**, **114** to achieve the particular frequency.

In a particular embodiment, the frequency and configuration correlation data maps the specific frequencies to particular deactivated radiating elements (e.g., the radiating elements **132** and **136**), to particular activated radiating elements (e.g., the radiating elements **130** and **134**), or to both. A particular radiating element (e.g., the radiating element **132**) may affect the frequency of operation of the antenna **100** differently than another radiating element (e.g., the radiating element **130**) when activated. In a particular embodiment, the controllers **112**, **114** may receive the frequency and configuration correlation data from the RF transmit/receive front-end section **108**.

The method **300** also includes selecting a first light source of a plurality of light sources based on the configuration, at **304**. For example, in FIG. 1, the controller **112** may select the LED **152** and the controller **114** may select the LED **156** based on the particular number (e.g., 2) of deactivated radiating elements indicated by the configuration.

The method **300** further includes activating the first light source. The first light source, in response to being activated, may transmit an optical signal to a first active filter node of a plurality of active filter nodes of the antenna. Each of a plurality of radiating elements of the antenna may be coupled to a particular active filter node of the plurality of active filter nodes. The first active filter node may deactivate a corresponding radiating element of the plurality of radi-

ating elements in response to the optical signal. For example, in FIG. 1, the controller 112 may activate the LED 152 and the controller 114 may activate the LED 156. The LED 152 may transmit the photonic signal 192 to the active filter node 122 and the LED 156 may transmit the photonic signal 196 to the active filter node 126. The active filter node 122 may deactivate the radiating element 132 in response to receiving the photonic signal 192 and the active filter node 126 may deactivate the radiating element 136 in response to receiving the photonic signal 196.

Thus, the method 300 may enable optical tuning of the antenna 100 to the particular frequency.

FIG. 4 is a block diagram illustrates an embodiment of a general computer system that is generally designated 400. The computer system 400 may be operable to support embodiments of computer-implemented methods, computer program products, and system components as illustrated in FIGS. 1-3. The computer system 400, or any portion thereof, may operate as a standalone device or may be activated, e.g., using a network, to other computer systems or peripheral devices.

The computer system 400 may be incorporated into a mobile computing device. The mobile computing device may include the antenna 100 of FIG. 1. The antenna 100 may be controlled by a controller 428 of the computer system 400.

The computer system 400 can also be implemented as or incorporated into various devices, such as a tablet PC, a personal digital assistant (PDA), a palmtop computer, a laptop computer, a communications device, a web appliance, a display device, a computing device, or any other machine capable of executing a set of instructions (sequential or otherwise) that specify actions to be taken by that machine. Further, while a single computer system 400 is illustrated, the term "system" shall also be taken to include any collection of systems or sub-systems that individually or jointly execute a set, or multiple sets, of instructions to perform one or more computer functions.

As illustrated in FIG. 4, the computer system 400 may include a processor 402, e.g., a central processing unit (CPU). In a particular embodiment, the processor 402 may include multiple processors. For example, the processor 402 may include distributed processors, parallel processors, or both. The multiple processors may be included in, or coupled to, a single device or multiple devices. The processor 402 may be used to support a virtual processing environment. In a particular embodiment, the processor 402 may include a state machine, an application specific integrated circuit (ASIC), or a programmable gate array (PGA) (e.g., a field programmable gate array (FPGA)).

Moreover, the computer system 400 may include a main memory 404 and a static memory 406 that may communicate with each other via a bus 408. In a particular embodiment, the main memory 404 may include processor-executable instructions 424. As shown, the computer system 400 may further include or be coupled to a display unit 410, such as a liquid crystal display (LCD), an organic light emitting diode (OLED), a flat panel display, a solid-state display, or a projection display. Additionally, the computer system 400 may include an input device 412, such as a keyboard, a remote control device, and a cursor control device 414, such as a mouse. In a particular embodiment, the cursor control device 414 may be incorporated into the remote control device. The computer system 400 may also include a disk drive unit 416, a signal generation device 418, such as a

speaker, and a network interface device 420. The network interface device 420 may be coupled to other devices (not shown) via a network 426.

In a particular embodiment, one or more of the components of the computer system 400 may be coupled to an antenna or an antenna system (e.g. the antenna 100 or a system coupled to the antenna 100 of FIG. 1). In a particular embodiment, the controller 428 may correspond to the controller 112 of FIG. 1, the controller 114 of FIG. 1, or both.

In a particular embodiment, as depicted in FIG. 4, the disk drive unit 416 may include a tangible computer-readable storage device 422 in which one or more sets of instructions 424, e.g. software, may be embedded. Further, the instructions 424 may embody one or more of the methods or logic as described herein. The processor 402 may execute the instructions 424 to perform operations corresponding to one or more of the methods or logic as described herein. The processor 402 may perform the operations directly, or the processor 402 may facilitate, direct, or cooperate with another device or component to perform the operations.

In a particular embodiment, the instructions 424 may reside completely, or at least partially, within the main memory 404, the static memory 406, and/or within the processor 402 during execution by the computer system 400. The main memory 404 and the processor 402 also may include tangible computer-readable media.

In an alternative embodiment, dedicated hardware implementations, such as application specific integrated circuits, programmable logic arrays and other hardware devices, can be constructed to implement one or more of the methods described herein. Applications that may include the apparatus and systems of various embodiments can broadly include a variety of electronic and computer systems. One or more embodiments described herein may implement functions using two or more specific interconnected hardware modules or devices with related control, or as portions of an application-specific integrated circuit. Accordingly, the present system encompasses software, firmware, and hardware implementations.

In accordance with various embodiments of the present disclosure, the methods described herein may be implemented by software programs executable by a computer system. Further, in an exemplary, non-limiting embodiment, implementations can include distributed processing and parallel processing. Alternatively, virtual computer system processing can be used to implement one or more of the methods or functionality as described herein.

The present disclosure describes a computer-readable non-transitory medium that includes instructions 424 so that an antenna may be optically tuned. Further, the instructions 424 may be transmitted or received over the network 426 via the network interface device 420 (e.g., via uploading and/or downloading of an optical antenna tuning application or program, or both).

While the computer-readable non-transitory medium is shown to be a single medium, the term "computer-readable medium" includes a single medium or multiple media, such as a centralized or distributed database, and/or associated caches and servers that store one or more sets of instructions. The term "non-transitory computer-readable medium" shall also include any medium that is capable of storing a set of instructions for execution by a processor or that cause a computer system to perform any one or more of the methods or operations disclosed herein.

In a particular non-limiting, exemplary embodiment, the computer-readable non-transitory medium can include a

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solid-state memory such as a memory card or other package that houses one or more non-volatile read-only memories. Further, the computer-readable non-transitory medium can be a random access memory or other volatile re-writable memory. Additionally, the computer-readable non-transitory medium can include a magneto-optical or optical medium, such as a disk or tapes. Accordingly, the disclosure is considered to include any one or more of a computer-readable non-transitory storage medium and successor media, in which data or instructions may be stored.

It should also be noted that software that implements the disclosed methods may optionally be stored on a tangible storage medium, such as: a magnetic medium, such as a disk or tape; a magneto-optical or optical medium, such as a disk; or a solid state medium, such as a memory card or other package that houses one or more read-only (non-volatile) memories, random access memories, or other re-writable (volatile) memories.

Although the present specification describes components and functions that may be implemented in particular embodiments with reference to particular standards and protocols, the claims are not limited to such standards and protocols. For example, standards for Internet, other packet switched network transmission and standards for viewing media content represent examples of the state of the art. Such standards are periodically superseded by faster or more efficient equivalents having essentially the same functions. Accordingly, replacement standards and protocols having the same or similar functions as those disclosed herein are considered equivalents thereof.

Moreover, although specific embodiments have been illustrated and described herein, it should be appreciated that any subsequent arrangement designed to achieve the same or similar purpose may be substituted for the specific embodiments shown. This disclosure is intended to cover any and all subsequent adaptations or variations of various embodiments. Combinations of the above embodiments, and other embodiments not specifically described herein, will be apparent to those of skill in the art upon reviewing the description.

The Abstract of the Disclosure is submitted with the understanding that it will not be used to interpret or limit the scope or meaning of the claims. In addition, in the foregoing Detailed Description, various features may be grouped together or described in a single embodiment for the purpose of streamlining the disclosure. This disclosure is not to be interpreted as reflecting an intention that the claimed embodiments require more features than are expressly recited in each claim. As the following claims reflect, inventive subject matter may be directed to less than all of the features of any of the disclosed embodiments. Thus, the following claims are incorporated into the Detailed Description, with each claim standing on its own as defining separately claimed subject matter.

The above-disclosed subject matter is to be considered illustrative, and not restrictive, and the appended claims are intended to cover all such modifications, enhancements, and other embodiments, which fall within the scope of the present disclosure. Thus, to the maximum extent allowed by law, the scope of the present disclosure is to be determined by the broadest permissible interpretation of the following claims and their equivalents, and shall not be restricted or limited by the foregoing detailed description.

What is claimed is:

1. An antenna system comprising:
a dielectric layer;

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- a first filter node coupled to a first surface of the dielectric layer;
- a second filter node coupled to the first surface of the dielectric layer;
- a first radiating element coupled to the first filter node and to the second filter node;
- a second radiating element coupled to the second filter node;
- an optical fiber coupled to the first filter node and to the second filter node, the optical fiber configured to carry a first light signal to power the first filter node and the second filter node;
- a first light source embedded in a ground layer located adjacent to a second surface of the dielectric layer, the first light source configured to transmit a second light signal to the second filter node to cause the second filter node to adjust a characteristic of the second radiating element using a portion of the power supplied by the first light signal, wherein a thickness of the dielectric layer contributes to a wavelength separation between the first light source and the second filter node, the wavelength separation being greater than a threshold distance to reduce any effect of the first light source on a field of the antenna; and
- a controller coupled to the first light source, the controller configured to activate the first light source to enable, or disable, use of the second radiating element as part of an antenna at a particular operating frequency based on the second light signal.

2. The antenna system of claim 1, wherein the second light signal produced by the first light source causes the second filter node to enable use of the second radiating element coupled to the second filter node.

3. The antenna system of claim 1, wherein the second light signal produced by the first light source causes the second filter node to disable use of the second radiating element coupled to the second filter node.

4. The antenna system of claim 1, further comprising a second light source configured to transmit a third light signal to the first filter node to enable use of the first radiating element and the second radiating element.

5. The antenna system of claim 1, further comprising a second light source configured to transmit a third light signal to the first filter node to disable use of the first radiating element and the second radiating element.

6. The antenna system of claim 1, wherein the second filter node comprises an inductance, a capacitance, a gain, a reactance, or a combination thereof.

7. The antenna system of claim 1, wherein the second filter node comprises an amplifier.

8. The antenna system of claim 7, wherein the second light signal is received by a switch of the second filter node, and wherein the switch is coupled to control the amplifier.

9. The antenna system of claim 8, wherein the amplifier of the second filter node uses the power from the first light signal for enabling or disabling the second radiating element.

10. An antenna system comprising:

- a dielectric layer;
- a first filter node coupled to a first surface of the dielectric layer;
- a second filter node coupled to the first surface of the dielectric layer;
- a first radiating element coupled to the first filter node and to the second filter node;
- a second radiating element coupled to the second filter node;

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- an optical fiber coupled to the first filter node and to the second filter node, the optical fiber configured to carry a power light signal to power the first filter node and the second filter node;
- a first light source configured to transmit a first light signal to the first filter node to cause the first filter node to enable or disable both the first radiating element and the second radiating element;
- a second light source configured to transmit a second light signal to the second filter node to cause the second filter node to enable or disable the second radiating element;
- a processing system including a processor coupled to the first light source and the second light source; and
- a memory that stores executable instructions that, when executed by the processing system, facilitate performance of operations, the operations comprising:
- determining a configuration of the antenna system based on an operating frequency; and
 - activating at least one of the first light source and the second light source based to place the antenna system into the configuration.
11. The antenna system of claim 10, wherein the second filter node comprises an inductance, a capacitance, a gain, a reactance, or a combination thereof.
12. The antenna system of claim 10, wherein the second filter node comprises an amplifier.
13. The antenna system of claim 12, wherein the second light signal is received by a switch of the second filter node, and wherein the switch is coupled to control the amplifier.
14. The antenna system of claim 12, wherein the amplifier of the second filter node uses the power from the power light signal for enabling or disabling the second radiating element.
15. An antenna system comprising:
- a dielectric layer;
 - a plurality of filter nodes coupled to a first surface of the dielectric layer;

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- a plurality of radiating elements coupled between pairs of filter nodes of the plurality of filter nodes;
 - an optical fiber coupled to the plurality of filter nodes, the optical fiber configured to carry a power light signal to power the plurality of filter nodes;
 - a plurality of light sources configured to transmit respective light signals to respective ones of the plurality of filter nodes to cause the respective ones of the plurality of filter nodes to enable or disable respective ones of the plurality of radiating elements;
 - a processing system including a processor coupled to the plurality of light sources; and
 - a memory that stores executable instructions that, when executed by the processing system, facilitate performance of operations, the operations comprising:
- determining a configuration of the antenna system based on an operating frequency; and
 - activating at least one of the plurality of light sources to place the antenna system into the configuration.
16. The antenna system of claim 15, wherein each filter node of the plurality of filter nodes comprises an inductance, a capacitance, a gain, a reactance, or a combination thereof.
17. The antenna system of claim 15, wherein each filter node of the plurality of filter nodes comprises an amplifier.
18. The antenna system of claim 17, wherein each filter node of the plurality of filter nodes comprises a switch, and wherein the respective light signals are received by the switch, and wherein the switch is coupled to control the amplifier.
19. The antenna system of claim 17, wherein the amplifier uses the power from the power light signal for enabling or disabling the respective ones of the plurality of radiating elements.
20. The antenna system of claim 17, wherein the activating the at least one of the plurality of light sources comprises disabling at least one of the plurality of radiating elements to shorten a length of the antenna system.

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